Algebraic extinction in acyclic spreading models: explicit counterexamples, genericity and support-checkable repairs

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Abstract

Chan, Crommelin and Mandjes conjectured that, in their class of interaction-driven spreading models, every vanishing compartment decays exponentially whenever two graphs built from the model are acyclic or nearly so. An earlier Evidence Press release refuted this conjecture with a four-compartment model decaying like $18/t^2$, proved four compartments necessary, and gave a spectral repair; within the source's encoding of recovery, a no-direct-infection SWIR-type submodel also admits a counterexample. The mechanism is an endgame in which the effective reproduction number of the active compartment tends to exactly one. This paper adds the following. A four-compartment model in which *both* graphs are acyclic has an explicit trajectory with $x(t) \sim \sqrt{3/8}\, t^{-3/2}$. With four compartments no tail is slower than $t^{-3/2}$. Acyclic chains give tails $t^{-1-1/m}$, optimal within models driven by one active catalyst. In those models algebraic extinction needs initial data in a null set, and exponential decay returns after a perturbation at a rate governed by a general slowing law. In the chain, SWIR and hierarchy families the slow tails never follow an outbreak peak. Conditions checkable from the coefficient supports or the rates restore the conjectured conclusion, and an exploratory simulation shows how finite populations cut the tail short.

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1 The question in words

Goal: say what is being asked, why the answer is not obvious, and what this paper proves.

The setting. In a compartmental model of an epidemic or a rumour, a closed population is divided into compartments: for example, people who have not heard a rumour, people who have heard it once, active spreaders, and people who have stopped spreading. Individuals move between compartments only when they meet someone; the person met (the *catalyst* of the move) keeps their own state. Chan, Crommelin and Mandjes [1] call this class *interaction-driven dispersion models*. It contains the classical SIR model of Kermack and McKendrick [3] and the Daley–Kendall and Maki–Thompson rumour models [6, 7]; spontaneous recovery is included by writing a recovery rate γx as $\gamma x \sum_j y_j$, a "contact with anyone" [1, eq. (8), p. 9].

Two directed graphs summarise a model. The *transition graph* records who can move where; the *dependency graph* records whose presence a compartment needs in order to empty. If the transition graph has no cycles, every compartment settles to a limit. Some compartments, such as the active spreaders, *vanish*: their fraction tends to zero. The question is *how fast*. For SIR the answer is exponential: the number of infectious people eventually halves at a fixed rate. Chan,

Crommelin and Mandjes proved that under their graph conditions a vanishing compartment is integrable, which rules out a $1/t$ tail, and they conjectured that the decay is always exponential.

Why the answer matters, and the picture behind it. Exponential fade-out is what modellers assume when they extrapolate a declining incidence curve to an elimination date. In SIR it holds for a simple reason. At the end of an outbreak the susceptible fraction has *overshot* the herd-immunity threshold, so the effective reproduction number $R_{\mathrm{eff}}(\infty)$ is strictly below one, and the infectious fraction decays like $\exp\{-(1-R_{\mathrm{eff}}(\infty))\gamma t\}$ [4, 5]. Algebraic extinction is what happens if the endgame lands *exactly* on $R_{\mathrm{eff}} = 1$. Then the active compartment is replenished as fast as it is removed, at leading order, and decay proceeds only through slower, higher-order effects. In our main example $R_{\mathrm{eff}}(t)$ tends to one from below,

$$1 - R_{\mathrm{eff}}(t) \sim \frac{9}{2t},$$

and the activity decays like $t^{-3/2}$ (Section 3). The graphs cannot see this balance, because it is a statement about limiting *rates*, not about which contacts are possible.

What was already known. Conjecture 1 has already been refuted. An earlier Evidence Press release [2] (by the same publisher as this paper) gives a four-compartment model with $I(t) \sim 18/t^2$, whose dependency graph contains the cycle $Y \to I \to Y$; it also proves that four compartments are necessary, constructs $(m+2)$-compartment models with tails $t^{-m/(m-1)}$, classifies its four-compartment model, and proves that the logarithmic decay rate of the vanishing mass equals the spectral bound of the limiting transverse Jacobian. Section 9 states precisely where the present paper overlaps with that release. Separately, the mean-field generalised epidemic process of Janssen, Müller and Stenull [9, Sec. II], called the SWIR model by Choi, Lee and Kahng [14], adds a "weakened" stage between susceptible and infectious. With recovery written as in the source, its no-direct-infection submodel belongs to the conjectured class and admits a tricritical-type triple zero, hence a $t^{-3/2}$ counterexample (Section 3.4). Its dependency graph has cycles, and its trajectory is not explicit. The source itself notes that integrable tails such as $1/t^2$ are not excluded by its integrability theorem [1, p. 24].

What this paper adds. Table 1 lists the results, marking those that overlap with [2], where to find them, and how each is established. The labels mean: *proof*, a complete written argument; *exact*, a symbolic or exact-rational computation reproduced by the package; *numerical*, a floating-point check; *exploratory*, a simulation or heuristic without proof; *open*, not settled.

Roadmap. Section 2 sets out the model, quotes the source's hypotheses verbatim, and introduces the two clocks used throughout. Section 3 derives the explicit counterexample, checks it against each hypothesis, and gives the t^{-2} and SWIR examples. Section 4 proves minimality. Section 5 gives the hierarchy, the dimension bound and the slowest four-compartment tail. Section 6 analyses the four-state chain completely: classification, genericity, the slowing law and the absence of a peak. Section 7 gives the limiting-block criterion and the support- and rate-checkable repairs. Section 8 discusses finite populations, and Section 9 the literature and open problems. Readers who want only the disproof need Sections 2 and 3.

Notation. Table 2 fixes the symbols; each is used in one sense only.

Result	Content	Status
SWIR in class (§3.4)	A no-direct-infection SWIR/GEP submodel admits a tricritical-type triple zero, hence a $t^{-3/2}$ counterexample.	proof; exact
Theorem 3.1	Four-state model, both graphs acyclic, explicit trajectory, $x \sim \sqrt{3/8}\, t^{-3/2}$; derived, not guessed.	proof; exact
Example (9)	Same graphs, $x \sim 4t^{-2}$, codimension one.	proof; exact
Theorem 4.1	Three compartments never suffice (as in [2], Thm 5).	proof
Theorem 5.1	Tails $t^{-1-1/m}$, every even m, both graphs acyclic; slower at each even N than [2], Thm 2.	proof; exact
Proposition 5.3	For $\dot{y} = xHy$: zero order $\leq 2\lfloor N/2 \rfloor - 1$, sharp.	proof
Corollary 5.4	At $N = 4$ with one vanishing state, $t^{-3/2}$ is the slowest tail.	proof
Theorem 5.7	Any $N = 4$ model: exponential, t^{-2} or $t^{-3/2}$; nothing slower.	proof; numerical examples
Theorem 6.1	All rates and data of the four-state chain: exponential, t^{-2} or $t^{-3/2}$.	proof
Theorem 6.2	Algebraic extinction is confined to a null set of data for $\dot{y} = xHy$; codimensions 1 and 2 in the chain.	proof; exact
Theorem 6.3	Order-k zero: perturbed rate $\propto \lvert\varepsilon\rvert^{(k-1)/k}$, endpoint shift $\propto \lvert\varepsilon\rvert^{1/k}$, jump for even k.	proof; numerical
Proposition 6.5	No outbreak peak precedes an algebraic tail in the chain, SWIR and hierarchy families.	proof; exact
Theorem 7.1	Logarithmic rate equals $s(J)$ of the limiting Metzler block (same statement as [2], Thm 7, Cor. 8).	proof
Corollary 7.3	Support-checkable condition for exponential extinction.	proof
Corollary 7.4	Rate-checkable condition; rates alone cannot decide a trajectory's tail.	proof
Section 8	Finite populations: extinction at $\approx 1.5N^{0.42}$.	exploratory

Table 1: Results and their status.

2 Model, the source's hypotheses, and two clocks

Goal: state the class exactly as the source does, so that every later verification can be checked line by line.

2.1 The model

There are N fractions $y_i > 0$ with $\sum_i y_i = 1$. For a donor i, a destination $j \neq i$ and a catalyst k (which may equal i or j) let $c_{ijk} \geq 0$ be constant. The flux $\varphi_{ijk} = c_{ijk} y_i y_k$ moves mass from i to j, and

$$\dot{y}_i = \sum_{a \neq i} \sum_k c_{aik}\, y_a y_k - y_i \sum_{b \neq i} \sum_k c_{ibk}\, y_k. \tag{1}$$

In the source notation $c_{ijk} = \beta^j_{ik}$ for $j \neq i$ and $\beta^i_{ik} = \sum_{j \neq i} c_{ijk}$, which is the source's conservation law [1, eq. (2), p. 7]; (1) is then the source's system [1, eqs. (1)–(4), pp. 6–7]. The simplex is invariant, $\dot{y}_i \geq -C_i y_i$ for a constant C_i, so positive coordinates stay positive at finite times, and solutions are global.

2.2 The hypotheses, quoted

Page numbers refer to the PDF of arXiv:2511.02437v1 (only version 1 exists at the time of writing); the file's SHA-256 is recorded in the package.

Symbol	Meaning
N, y_i, y^*	number of compartments; fraction in compartment i; limit of $y(t)$
c_{ijk}, β_{ik}^j	contact coefficient (donor i, destination j, catalyst k); source notation, $c_{ijk} = \beta_{ik}^j$
φ_{ijk}	flux $c_{ijk} y_i y_k$
$\mathcal{V}, \mathcal{P}$	vanishing and persistent compartments ($y_i^* = 0$, resp. > 0)
x	fraction in the active compartment X
$u(t)$, U	exposure clock $\int_0^t x$; its limit $u(\infty)$
$\Phi(u)$	active fraction as a function of exposure, $x(t) = \Phi(u(t))$
s, w, x, r	fractions in the base chain $S \to W \to X \to R$
$\vartheta(t)$, $\zeta(t)$	$\vartheta = (16 + 2t/3)^{-1/2}$, $\zeta = 1 + \vartheta$ (base chain)
α, β, γ	rates of the general four-state chain
q, $Q(q)$, q_*	$q = e^{-u}$; exposure cubic for rates $(2, 1, 1)$; its attained root
κ	second exposure derivative at a double zero
ν, η	SWIR rates $S \to W$ and $W \to I$ (recovery rate 1)
m, w_k, ϱ	hierarchy order, reservoir fractions, $\varrho = 1/(m+1)$
H, k, a_k	matrix in $\dot{y} = xHy$; order of the first zero of Φ; $\Phi(U - v) = a_k v^k + \dots$
M_N	$2\lfloor N/2 \rfloor - 1$
J, $s(J)$	limiting Metzler block of the vanishing coordinates; spectral abscissa
ε, δ, Γ	perturbation size and direction; induced profile change $\Gamma(u)$
ω	an exponential decay rate

Table 2: Notation.

- Initial data [1, p. 6]: "Initial conditions are assumed to be $y_i(0) \in (0, 1)$."
- Transition graph, Definition 6 [1, p. 7]: "The set of directed edges E contains an edge $(u, v) \in V \times V$ with $u \neq v$ if and only if there exists some state $k \in V$ such that $\beta_{uk}^v > 0$."
- Dependency graph, Definition 7 [1, p. 8]: "The set of directed edges E contains an edge $(u, v) \in V \times V$ if and only if $\beta_{uv}^u > 0$." And: "(Self-loops $i \to i$ occur when $\beta_{ii}^i > 0$.)"
- Assumption 1 [1, p. 11]: "The underlying transition graph of the interaction-driven dispersion model is a DAG."
- Definition 8 [1, p. 14]: "We call state i vanishing if $\lim_{t\to\infty} y_i(t) = 0$ and persistent if $\lim_{t\to\infty} y_i(t) > 0$."
- Assumption 2 [1, p. 15]: "Every cycle in the underlying dependency graph contains at least one state with a persistent dependency successor."
- Meaning of exponential decay [1, §3.3, p. 24]: "there exist constants $C > 0$ and $r > 0$ such that $y_i(t) \leq Ce^{-rt}$."
- Conjecture 1 [1, p. 25]: "Under Assumption 1 and 2, every vanishing state decays exponentially to 0."

In our notation: the transition graph has an edge $i \to j$ iff $c_{ijk} > 0$ for some k; the dependency graph has an edge $i \to k$ iff $c_{ijk} > 0$ for some j (donor to catalyst, self-loops allowed when $k = i$). We write (G1) for Assumption 1 and (G2) for Assumption 2. Persistence is judged by limits, as in Definition 8, so (G2) refers to labels of the trajectory. The source's Corollary 1 [1, p. 22] reads: "Under Assumptions 1 and 2, if the persistent successor within each cycle of the dependency graph is identifiable, then the asymptotic characterization of every state can be determined." It adds that this characterization depends "only on whether each β-coefficient is zero or non-zero". So, when that identifiability condition holds, the labels are determined by the supports of the coefficients. The conjecture has no genericity clause: one trajectory suffices to refute it. Our explicit counterexamples have acyclic graphs without self-loops under either orientation of the dependency edges, so they do not depend on this convention. The minimality theorem (Section 4)

uses the orientation above, which is the source's.

2.3 Two facts and two clocks

Lemma 2.1 (Finite total flux under (G1))**.** *Under (G1), every coordinate has a limit y_i^*, every flux is integrable on $[0,\infty)$ and tends to zero, and*

$$c_{ijk} > 0 \implies y_i^* y_k^* = 0. \tag{2}$$

Proof. Label compartments topologically, so $c_{ijk} > 0$ implies $i < j$. The bounded function $L = \sum_i i y_i$ has $\dot{L} = \sum (j-i)\varphi_{ijk} \geq \sum \varphi_{ijk} \geq 0$, because $j - i \geq 1$. Hence the total flux is integrable, so is each $|\dot{y}_i|$, and each coordinate converges. Each flux then has a limit, which is zero because the flux is integrable. □

Lemma 2.2 (Integrability and persistent successors)**.** *Under (G1)–(G2), every vanishing coordinate is integrable and has a persistent dependency successor, and every dependency successor of a persistent state vanishes.*

Proof. A state i with a persistent dependency successor k is integrable: some outflow $c_{ijk} y_i y_k$ is eventually at least $\frac{1}{2} c_{ijk} y_k^* \, y_i$ and is integrable by Lemma 2.1. Let B be the set of vanishing, non-integrable states and suppose $B \neq \varnothing$. By the previous sentence no member of B has a persistent dependency successor. Since $y_i(t) \geq y_i(0) \exp(-\int_0^t h_i)$ with per-capita outflow $h_i = \sum_{j,k} c_{ijk} y_k$, a vanishing i has $\int_0^\infty h_i = \infty$, so some dependency successor of i is non-integrable. That successor is not persistent (persistent states are not integrable), and it is not integrable, so it lies in B. Following successors inside the finite set B produces a cycle in B, and (G2) gives one of its vertices a persistent successor: a contradiction. Thus $B = \varnothing$. If a vanishing state had only vanishing successors, all would be integrable, its integrated hazard would be finite, and it could not vanish. Finally a persistent state cannot have a persistent successor by (2). □

Both lemmas restate results of the source [1, Theorem 2, p. 17; Theorem 3, p. 19]; we include proofs because later arguments use their internal steps.

Two clocks. Ordinary time t is the clock in which we measure decay. The *exposure clock* is the cumulative activity

$$u(t) = \int_0^t x(\sigma)\, \mathrm{d}\sigma,$$

the total amount of contact with the active compartment up to time t. When every flux has the active fraction x as a factor, dividing by $\dot{u} = x$ makes the equations *linear* in u; this is the device behind Kermack and McKendrick's final-size equation [3] and its use in rumour models [24, 25]. In exposure time everything is explicit. The only nonlinear step is to convert back to t through the scalar equation $\dot{u} = \Phi(u)$, where $\Phi(u)$ is the active fraction at exposure u. The total exposure $U = u(\infty)$ is finite exactly when x is integrable, and then $x(t) \to 0$ at a rate set by how Φ vanishes at U: a simple zero gives exponential decay, and a zero of order $k \geq 2$ gives $t^{-k/(k-1)}$. Exponential decay versus a power law is therefore a question about the multiplicity of one zero of an explicit function. In dynamical-systems terms the endpoint is a non-hyperbolic boundary equilibrium, and the slow tail is an approach along a centre direction [20]; what is at issue is whether that approach is compatible with the source's graph conditions.

3 The counterexample, derived

Goal: construct an explicit trajectory that satisfies every hypothesis of the conjecture and still decays like $t^{-3/2}$, showing where the numbers come from.

3.1 A staged rumour with stifling

Take four compartments: S (unaware), W (primed: heard once), X (active spreader) and R (stifler), with fractions s, w, x, r. There are three contact moves, shown in Figure 1: an unaware person who meets a spreader becomes primed (rate 2); a primed person who meets a spreader becomes a spreader (rate 1); a spreader who meets a stifler stops spreading (rate 1). In the source's notation the only nonzero coefficients are

$$\beta_{SX}^{W} = 2, \qquad \beta_{WX}^{X} = 1, \qquad \beta_{XR}^{R} = 1,$$

and the equations are

$$\dot{s} = -2sx, \qquad \dot{w} = 2sx - wx, \qquad \dot{x} = (w - r)x, \qquad \dot{r} = rx. \tag{3}$$

This is a two-step version of the rumour-scotching model of de Arruda et al. [24, eq. (3.2)], close to the repetition models of Rada et al. [25]. In this model the effective reproduction number of the spreaders is $R_{\text{eff}} = w/r$, the ratio of their per-capita production and removal rates.

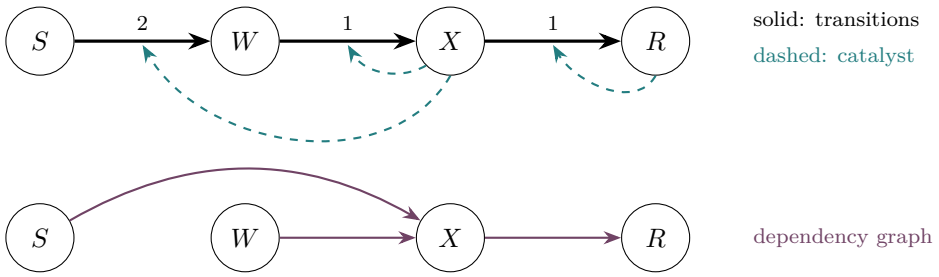


Figure 1: The base chain. Top: transition graph (solid) with the catalyst of each move (dashed); the transition graph is the path $S \to W \to X \to R$. Bottom: the dependency graph, with edges from each donor to the catalyst of its outflow. Both graphs are acyclic and have no self-loops. Note that $W \to X$ is catalysed by its own destination X; this autocatalytic inflow is invisible to the dependency graph and is the source of the slow tail (Section 7).

3.2 Finding the trajectory

Every flux in (3) contains x, so in the exposure clock u the system is linear: $s' = -2s$, $w' = 2s - w$, $r' = r$ (primes denote $\mathrm{d}/\mathrm{d}u$). With $q = e^{-u}$ and initial data (s_0, w_0, x_0, r_0),

$$s = s_0 q^2, \qquad w = (w_0 + 2s_0)q - 2s_0 q^2, \qquad r = \frac{r_0}{q}, \qquad x = 1 - s - w - r = \frac{Q(q)}{q},$$

with the *exposure cubic*

$$Q(q) = s_0 q^3 - (w_0 + 2s_0)q^2 + q - r_0. \tag{4}$$

Since $\dot{q} = -qx = -Q(q)$, the activity vanishes when q reaches the largest root $q_* \in (0, 1)$ of Q (note $Q(0) = -r_0 < 0 < x_0 = Q(1)$). A simple root gives exponential decay. To make the decay as slow as possible we ask for a triple root, $Q(q) = s_0(q - q_*)^3$. Matching coefficients gives

$$s_0 = \frac{1}{3q_*^2}, \qquad w_0 = \frac{3q_* - 2}{3q_*^2}, \qquad r_0 = \frac{q_*}{3}, \qquad x_0 = \frac{(1 - q_*)^3}{3q_*^2}, \tag{5}$$

which is positive exactly for $2/3 < q_* < 1$. Then $\dot{q} = -s_0(q - q_*)^3$, and $\vartheta = q/q_* - 1$ solves $\dot{\vartheta} = -\vartheta^3/3$. The choice $q_* = 4/5$ gives rational data and $\vartheta(0) = 1/4$.

Theorem 3.1 (Explicit counterexample with both graphs acyclic)**.** *Let* $\vartheta(t) = (16 + \frac{2}{3}t)^{-1/2}$ *and* $\zeta = 1 + \vartheta$*. The solution of* (3) *with*

$$(s_0, w_0, x_0, r_0) = \Big(\frac{25}{48}, \frac{5}{24}, \frac{1}{240}, \frac{4}{15}\Big) \tag{6}$$

is

$$s = \frac{\zeta^2}{3}, \qquad w = \frac{\zeta(3 - 2\zeta)}{3}, \qquad x = \frac{\vartheta^3}{3\zeta}, \qquad r = \frac{1}{3\zeta}. \tag{7}$$

It is strictly positive, tends to $(1/3, 1/3, 0, 1/3)$*, and*

$$x(t) \sim \sqrt{3/8}\, t^{-3/2}, \qquad \int_0^\infty x(t)\, \mathrm{d}t = \log\frac{5}{4}. \tag{8}$$

Every hypothesis of Conjecture 1 holds, so the conjecture is false even when the dependency graph is also acyclic.

Proof. With $q_* = 4/5$, (5) gives (6), and $\zeta = q/q_*$. Substituting $q = \frac{4}{5}\zeta$ in the exposure formulas gives (7); the identity $1 + \vartheta^3 = \zeta(\zeta^2 - 3\zeta + 3)$ confirms that the four fractions sum to one. Since $0 < \vartheta \le 1/4$ we have $1 < \zeta \le 5/4$ and $3 - 2\zeta \ge 1/2$, so all fractions are positive. As $t \to \infty$, $\vartheta \to 0$, which gives the limit, and $x \sim \frac{1}{3}(\frac{3}{2})^{3/2} t^{-3/2} = \sqrt{3/8}\, t^{-3/2}$. The total exposure is $u(\infty) = \log(1/q_*) = \log(5/4)$. A power law excludes any bound Ce^{-rt}, so x is a vanishing state without exponential decay. The hypothesis check is Table 3. □

Source hypothesis	Check for (3)–(6)
$y_i(0) \in (0, 1)$ (p. 6)	$25/48, 5/24, 1/240, 4/15$ lie in $(0, 1)$ and sum to one.
Model (1)–(4), conservation (2) (pp. 6–7)	Coefficients constant and nonnegative; $\beta^S_{SX} = \beta^W_{SX} = 2$, $\beta^W_{WX} = \beta^X_{WX} = 1$, $\beta^X_{XR} = \beta^R_{XR} = 1$, all others zero.
Def. 6 transition graph (p. 7)	Edges $S \to W$, $W \to X$, $X \to R$: a path.
Assumption 1 (p. 11)	The path is a DAG.
Def. 7 dependency graph (p. 8)	$\beta^u_{uv} > 0$ only for (S, X), (W, X), (X, R); no self-loops since $\beta^i_{ii} = 0$ for all i.
Assumption 2 (p. 15)	The dependency graph has no cycle, so the condition holds vacuously (the source notes it "is met by any DAG", p. 15).
Def. 8 labels (p. 14)	S, W, R persistent ($\to 1/3$), X vanishing.
Exponential decay (p. 24)	Fails: $x(t)\, e^{\epsilon t} \to \infty$ for every $\epsilon > 0$.

Table 3: Hypothesis-by-hypothesis check against the source, with page pinpoints from Section 2.2.

The balance in words. Along the trajectory $R_{\text{eff}} = w/r = \zeta^2(3 - 2\zeta)$, so

$$1 - R_{\text{eff}} = (\zeta - 1)^2(2\zeta + 1) \sim \frac{9}{2t}, \qquad \frac{\dot{x}}{x} = w - r = -r(1 - R_{\text{eff}}) \sim -\frac{3}{2t},$$

which is the exponent $3/2$. Heuristically, a tail $t^{-\theta}$ corresponds to $1 - R_{\text{eff}}(t) \approx \theta/(\gamma r^* t)$, where γr^* is the limiting per-capita removal rate.

How tuned is this? The data (6) are tuned: at fixed rates the $t^{-3/2}$ trajectories form the one-parameter curve (5) in the two-dimensional simplex of initial data, which has codimension two (Theorem 6.2). This does not weaken the disproof, because Conjecture 1 is universally quantified. It does mean that the right repair is a genericity statement, which we prove in Section 6. A less tuned counterexample exists with the same graphs.

3.3 A t^{-2} counterexample of codimension one

Asking only for a double root, $Q(q) = s_0(q-q_*)^2(q-q_\circ)$ with $q_\circ < q_*$, gives a two-parameter family (Section 6). For example,

$$(s_0, w_0, x_0, r_0) = \Big(\frac{50}{81}, \frac{25}{162}, \frac{11}{3240}, \frac{9}{40}\Big), \qquad Q(q) = \frac{(10q-9)^2(20q-9)}{3240}, \qquad x(t) \sim 4t^{-2}, \quad (9)$$

with endpoint $(1/2, 1/4, 0, 1/4)$ and total exposure $\log(10/9)$. These trajectories form a surface of codimension one in the space of initial data. A one-parameter sweep, for instance of the initial stifler fraction, generically crosses this surface at an isolated value, so a t^{-2} tail is met in one-parameter families without further tuning. It also refutes Conjecture 1.

3.4 A SWIR-type counterexample

The mean-field generalised epidemic process [9, Sec. II, eq. (2.3)] (the SWIR model in [14]), restricted to the submodel without direct infection, reads

$$\dot{s} = -\nu s x, \quad \dot{w} = \nu s x - \eta w x, \quad \dot{x} = \eta w x - x, \quad \dot{r} = x, \tag{10}$$

with x the infectious fraction and recovery rate one. Writing x as $x(s+w+x+r)$, as the source does for SIR [1, eq. (8), p. 9], puts (10) in the class. Its transition graph is the path $S \to W \to X \to R$; its dependency graph has edges $S \to X$, $W \to X$ and $X \to \{S, W, X, R\}$, so its cycles are $X \to X$, $S \to X \to S$ and $W \to X \to W$. Each contains X, which has the persistent successor R, so (G2) holds. In exposure time, $s = s_0 e^{-\nu u}$, $w = e^{-\eta u}(w_0 + \nu s_0 u)$ when $\nu = \eta$, and $r = r_0 + u$. The profile $\Phi = 1 - s - w - r$ satisfies $\Phi' = \eta w - 1$, $\Phi'' = \eta(\nu s - \eta w)$ and $\Phi''' = -\nu\eta$ at a point where the first two vanish. Hence the endpoint $(s^*, w^*, r^*) = (1/\nu, 1/\eta, 1 - 1/\nu - 1/\eta)$ is a triple zero, of the tricritical type (we do not identify it with the tricritical point analysed in [9]). With $\nu = \eta = 4$ and total exposure $1/10$, integrating backwards gives the interior data

$$(s_0, w_0, x_0, r_0) = \Big(\tfrac{1}{4}e^{2/5},\ \tfrac{3}{20}e^{2/5},\ \tfrac{1}{5}(3 - 2e^{2/5}),\ \tfrac{2}{5}\Big) \approx (0.3730, 0.2238, 0.00327, 0.4), \tag{11}$$

The profile is positive before $U = 1/10$: as in the proof of Proposition 6.5(b), Φ' has a double zero at U and at most two zeros, so $\Phi' < 0$ on $[0, U)$ and Φ decreases to $\Phi(U) = 0$. Every flux contains x, so the model has the form (15), and Proposition 5.3 (Step 4) with $a_3 = \nu\eta/6$ gives $x(t) \sim \frac{\sqrt{3}}{8} t^{-3/2}$ (exact and quadrature checks in the package). We do not know whether such an interior trajectory, or its bearing on Conjecture 1, appears in the generalised-epidemic literature [8, 9, 11, 14, 13], where the $t^{-3/2}$ law is discussed in the context of tricritical scaling [9, Sec. V]. The base chain differs from SWIR in three ways: both of its graphs are acyclic; removal is by contact with stiflers rather than spontaneous; and its trajectory is explicit.

4 Why three compartments cannot suffice

Goal: show that the four compartments of Theorem 3.1 (and of SWIR) are necessary.

Theorem 4.1 (Minimal number of compartments)**.** *If $N \le 3$, the initial data are interior and (G1)–(G2) hold, then every vanishing coordinate decays exponentially.*

Proof. Order compartments topologically. The last is a sink of the transition graph; it never loses mass and so persists. By Lemma 2.2 each vanishing state has a persistent dependency successor, so its per-capita outflow has a positive limit. A vanishing first compartment has no inflow, so it decays exponentially. If the first two compartments vanish, the second has inflow at

most a constant times the first and an outflow hazard bounded below, so variation of constants gives exponential decay (an extra factor t from equal rates is absorbed by lowering the rate slightly). If the first compartment persists and the second vanishes when $N = 2$, the sink is the second compartment, a contradiction; the case "first vanishing, second persistent" is covered by the source argument.

The remaining configuration is $N = 3$ with persistent first and third compartments and a vanishing middle one; write their fractions as (y_1, x, y_3). By (2) every outflow from compartment 1 is catalysed by x, so the most general equations are

$$\begin{aligned} \dot{y}_1 &= -(c_1 + c_3)y_1 x, \\ \dot{x} &= x\big[(c_1 - c_2)y_1 - c_5 y_3 - c_4 x\big], \\ \dot{y}_3 &= x\big[(c_2 + c_3)y_1 + c_5 y_3 + c_4 x\big], \end{aligned}$$

with $c_1, \dots, c_5 \geq 0$: c_1 is $1 \to 2$ catalysed by X, c_2 is $X \to 3$ catalysed by 1, c_3 is $1 \to 3$ catalysed by X, c_4 is $X \to 3$ catalysed by X (a self-loop in the dependency graph), and c_5 is $X \to 3$ catalysed by 3. The persistent-successor property of x gives $c_2 + c_5 > 0$. Put $\lambda_* = (c_1 - c_2)y_1^* - c_5 y_3^*$; since $\dot{x}/x \to \lambda_*$ and $x \to 0$, $\lambda_* \leq 0$. Suppose $\lambda_* = 0$. If $c_1 + c_3 = 0$ then $\lambda_* = -c_2 y_1^* - c_5 y_3^* < 0$; so $c_1 + c_3 > 0$ and $\int_0^\infty x = \log(y_1(0)/y_1^*)/(c_1 + c_3) < \infty$. If $c_5 > 0$, $\lambda_* = 0$ forces $c_1 > c_2$; as y_1 decreases and y_3 increases, $(c_1 - c_2)y_1(t) - c_5 y_3(t) \geq \lambda_* = 0$. If $c_5 = 0$, $\lambda_* = 0$ forces $c_1 = c_2$ and the expression vanishes. In both cases $\dot{x} \geq -c_4 x^2$, so $x(t) \geq x(0)/(1 + c_4 x(0)t)$, which is not integrable: a contradiction. Hence $\lambda_* < 0$ and x decays exponentially. □

5 Arbitrarily slow integrable tails, and a dimension bound

Goal: show that the exponent $3/2$ *is not a barrier: with more compartments the tail can be as close to* $1/t$ *as we like, and in a natural subclass we determine exactly how close at each* N*.*

5.1 The hierarchy

Fix an even $m \geq 2$. Compartments $W_m, \dots, W_1$ record how many more contacts with spreaders a person needs before becoming one; X and R are as before. With fractions $w_m, \dots, w_1, x, r$,

$$\dot{w}_k = x\big[(k+1)w_{k+1} - k w_k\big] \ (1 \leq k \leq m,\ w_{m+1} = 0), \qquad \dot{x} = x(w_1 - r), \qquad \dot{r} = rx. \tag{12}$$

The move $W_k \to W_{k-1}$ ($k \geq 2$) has flux $k w_k x$, $W_1 \to X$ has flux $w_1 x$, and $X \to R$ is catalysed by R. Both graphs are acyclic. The case $m = 2$ is (3) with $(w_2, w_1) = (s, w)$.

Theorem 5.1 (An explicit hierarchy with two acyclic graphs)**.** *Put* $\varrho = 1/(m+1)$, $\vartheta(t) = [(2m)^m + \frac{m}{m+1}t]^{-1/m}$ *and* $\zeta = 1 + \vartheta$*. Then*

$$w_k = \varrho\, \zeta^k \sum_{\ell=0}^{m-k} \binom{k+\ell}{k} (-\vartheta)^\ell, \qquad r = \frac{\varrho}{\zeta}, \qquad x = \frac{\varrho\, \vartheta^{m+1}}{\zeta} \tag{13}$$

solve (12)*, with positive rational initial fractions summing to one. Every* w_k *and* r *tend to* ϱ*, and*

$$x(t) \sim C_m t^{-1-1/m}, \qquad C_m = \Big(\frac{m+1}{m^{m+1}}\Big)^{1/m}, \qquad \int_0^\infty x\,\mathrm{d}t = \log\Big(1 + \frac{1}{2m}\Big). \tag{14}$$

Hence no $\epsilon > 0$ *gives* $x = O(t^{-1-\epsilon})$ *uniformly over all finite dimensions.*

Proof. Positivity. $\vartheta(0) = 1/(2m)$ and ϑ decreases to zero. For $k < m$ the alternating sum $S_k = \sum_\ell \binom{k+\ell}{k}(-\vartheta)^\ell$ has consecutive term ratios $\frac{k+\ell+1}{\ell+1}\vartheta \leq (k+1)\vartheta \leq \frac{1}{2}$, so pairing terms gives $S_k \geq 1 - (k+1)\vartheta \geq 1/2$; $S_m = 1$.

Reservoirs. Let $\mathcal{S}(\xi,\zeta)=\varrho\sum_{j=1}^{m}(1-\zeta+\zeta\xi)^j$. For $k\geq 1$ the coefficient of ξ^k is $\varrho\zeta^k\sum_{j\geq k}\binom{j}{k}(1-\zeta)^{j-k}=w_k$. The identity $-\zeta\partial_\zeta\mathcal{S}=(1-\xi)\partial_\xi\mathcal{S}$ gives, on equating coefficients, $\mathrm{d}w_k/\mathrm{d}u=(k+1)w_{k+1}-kw_k$ along any clock with $\mathrm{d}\zeta/\mathrm{d}u=-\zeta$. From the definition, $\dot\vartheta=-\varrho\vartheta^{m+1}=-\zeta x$, so $\dot\zeta=-\zeta x$ and the reservoir equations hold in t.

Conservation and the active equation. Setting $\xi=1$, $\sum_k w_k=\varrho\sum_{j=1}^{m}[1-(1-\zeta)^j]$. Since $(m+1)\varrho=1$,

$$1-\sum_k w_k-\frac{\varrho}{\zeta}=\varrho\sum_{j=0}^{m}(1-\zeta)^j-\frac{\varrho}{\zeta}=-\frac{\varrho(1-\zeta)^{m+1}}{\zeta}=\frac{\varrho\vartheta^{m+1}}{\zeta}=x,$$

where the last-but-one equality uses that m is even. Then $\dot r=rx$, and summing reservoir equations gives $\sum\dot w_k=-xw_1$, so conservation yields $\dot x=x(w_1-r)$.

Rate. $\vartheta\sim((m+1)/(mt))^{1/m}$ gives (14), and integrating $\dot\zeta/\zeta=-x$ gives the exposure. Given $\epsilon>0$, take even $m>1/\epsilon$. □

Remark 5.2 (Why m must be even). For odd m the same formulas give $x=-\varrho\vartheta^{m+1}/\zeta<0$: for $m=3$ the initial active fraction is $-1/6048$. Odd hierarchies are replaced by the terminal split in the proof of Proposition 5.3.

m	compartments	tail exponent	C_m	time scale $T_m=(m+1)(2m)^m/m$
2	4	3/2	$\sqrt{3/8}$	24
4	6	5/4	$(5/1024)^{1/4}$	5120
6	8	7/6	$(7/279936)^{1/6}$	$\approx 3.48\times10^7$

Table 4: Members of the hierarchy. T_m is the time after which the power law sets in: $\vartheta(t)=\vartheta(0)(1+t/T_m)^{-1/m}$.

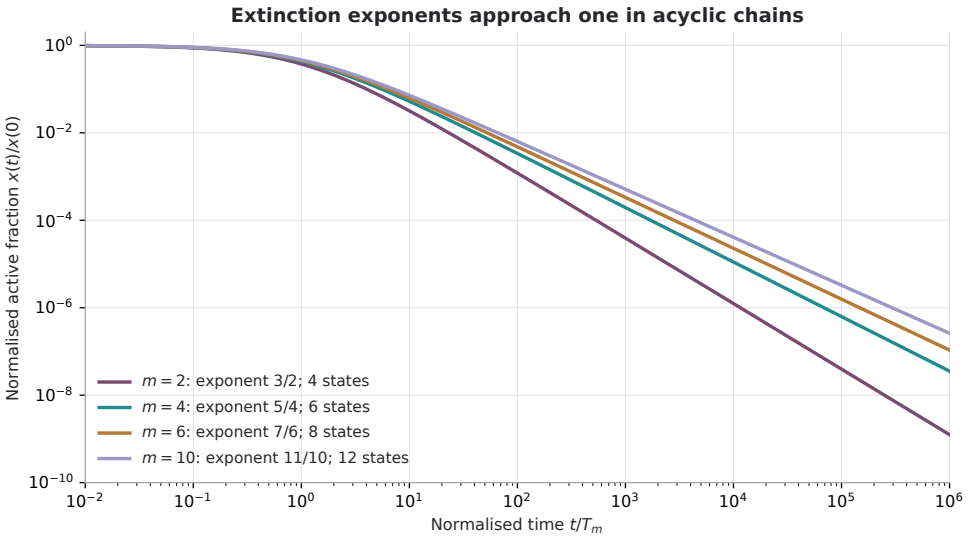


Figure 2: Hierarchy trajectories, activity normalised by its initial value and time by T_m. Slopes approach $-1-1/m$. Illustration only.

5.2 A sharp bound when one catalyst drives everything

Many models in this paper (the base chain, SWIR, the hierarchy) have a special property: every flux contains the active fraction x. Then the whole system has the global form

$$\dot y=x\,Hy,\qquad H\text{ constant},\tag{15}$$

as an identity of polynomial vector fields on the simplex, and the exposure clock linearises it globally: $y(t) = e^{Hu(t)}y(0)$ and $x(t) = \Phi(u(t))$ with $\Phi(u) = e_x^{\mathsf{T}} e^{Hu} y(0)$. Since $\mathbf{1}^{\mathsf{T}}\dot{y} = 0$ identically, $\mathbf{1}^{\mathsf{T}} H = 0$.

Proposition 5.3 (Sharp dimension and parity bound)**.** *Let (15) be an N-state model satisfying (G1)–(G2) with interior data and $x \to 0$, and let $U = \int_0^\infty x < \infty$. Then Φ has a zero at U of finite order k with $1 \le k \le M_N = 2\lfloor N/2 \rfloor - 1$. If $\Phi(U - v) = a_k v^k + O(v^{k+1})$ with $a_k > 0$, then $k = 1$ gives $x \sim Ce^{-a_1 t}$ for some $C > 0$, and $k \ge 2$ gives*

$$x(t) \sim a_k^{-1/(k-1)} (k-1)^{-k/(k-1)}\, t^{-k/(k-1)}. \tag{16}$$

For $N \ge 4$ the algebraic exponent is at least $M_N/(M_N - 1)$. This is attained in every $N \ge 4$ with positive rational data and two acyclic graphs; for odd N by duplicating the terminal compartment.

Proof. Step 1 (order at most $N - 1$). By Lemma 2.2, $U < \infty$; $\Phi > 0$ on $[0, U)$ and $\Phi(U) = 0$. By Cayley–Hamilton, Φ solves the order-N linear constant-coefficient equation given by the characteristic polynomial of H. If Φ and its first $N - 1$ derivatives vanished at U, then $\Phi \equiv 0$, contradicting $\Phi(0) = x_0 > 0$. So $1 \le k \le N - 1$, and $a_k > 0$ by positivity to the left of U.

Step 2 (structure). Every nonzero flux contains x: with the topological weight $L = \sum_i i y_i$, the identity (15) gives $\dot{L} = 0$ on the face $\{x = 0\}$, while the contact form gives $\dot{L} = \sum (j - i)\varphi_{ijk}$ with positive weights; at a point of the face with all other fractions positive, a flux without the factor x would make $\dot{L} > 0$. Write the other $n = N - 1$ coordinates as z, so $\mathrm{d}z/\,\mathrm{d}u = Bz + b\Phi(u)$ with B Metzler and $b \ge 0$ (a negative entry can only be a donor's own outflow). The transition graph has a sink $j \ne X$ (a sink X could not vanish), and $B_{jj} \ge 0$; hence $\sigma_B = s(B) \ge 0$.

Step 3 (parity). Suppose $k = n$. Let $z^* = z(U)$, $g(v) = \mathbf{1}^{\mathsf{T}} e^{Bv} z^*$ and $\chi(\lambda) = \det(\lambda I - B)$. Conservation and $\Phi(U) = 0$ give $g(0) = 1$. Since $\Phi^{(j)}(U) = 0$ for $j < n$, induction on j in $z^{(j+1)} = Bz^{(j)} + b\Phi^{(j)}$ gives $z^{(j)}(U) = B^j z^*$ for $j \le n$. Differentiating conservation then yields $g^{(j)}(0) = 0$ for $1 \le j < n$ and $\Phi^{(n)}(U) = -g^{(n)}(0)$; Cayley–Hamilton for B gives

$$\Phi^{(n)}(U) = \chi(0) \ne 0. \tag{17}$$

The spectral abscissa of a Metzler matrix is an eigenvalue, so $\chi(0) \ne 0$ and $\sigma_B \ge 0$ give $\sigma_B > 0$. As $e^{Bv} \ge 0$ and $z^* \ge 0$ has mass one, $g > 0$, and its Laplace transform converges and is positive for real $\lambda > \sigma_B$. Using $\chi(D)g = 0$ and the initial derivatives,

$$0 < \int_0^\infty e^{-\lambda v} g(v)\,\mathrm{d}v = \frac{\chi(\lambda) - \chi(0)}{\lambda \chi(\lambda)}, \qquad \lambda > \sigma_B. \tag{18}$$

As $\lambda \downarrow \sigma_B$, $\chi(\lambda) \downarrow 0$ through positive values, so the right side has the sign of $-\chi(0)$: hence $\chi(0) < 0$. Positivity of Φ before U requires $(-1)^n \Phi^{(n)}(U) > 0$, so with (17) n is odd. Thus $k = N - 1$ is impossible for odd N, and $k \le M_N$.

Step 4 (rates and sharpness). With $v = U - u$, $\dot{v} = -a_k v^k (1 + O(v))$; for $k = 1$ the logarithmic remainder is integrable, and for $k \ge 2$, $\mathrm{d}(v^{1-k})/\,\mathrm{d}t \to (k-1)a_k$, giving (16). For even $N = m + 2$ the hierarchy has $\Phi(u) = \varrho(e^{U-u} - 1)^{m+1} e^{-(U-u)}$, of order $m + 1 = M_N$. For odd $N \ge 5$, take the hierarchy with $N - 1$ states and split R into R_1, R_2 with $r_1 = r_2 = r/2$, replacing $X \to R$ by the four moves $X \to R_i$ catalysed by R_j with flux $\frac{1}{2} x r_j$. Then $\dot{r}_i = \frac{1}{2}(r_1 + r_2)x$, the active trajectory is unchanged, the graphs stay acyclic, and the order is $N - 2 = M_N$. □

The Cayley–Hamilton bound in Step 1 is standard; the parity step is what uses positivity, acyclicity and conservation. The odd-dimensional sharpness is obtained by duplication and adds no new dynamics. The proposition concerns only the class (15).

Corollary 5.4 (Slowest tail at $N = 4$ with one vanishing state)**.** *Let $N = 4$, (G1)–(G2) hold, the data be interior, and exactly one compartment vanish. Then that compartment is $O(t^{-3/2})$, and either it decays exponentially or like t^{-2} or $t^{-3/2}$. The bound is attained by Theorem 3.1.*

Proof. If only X vanishes, (2) forces every nonzero flux to contain x, so the model has the form (15). Proposition 5.3 gives $k \le M_4 = 3$. □

5.3 The slowest tail with four compartments

Goal: show that at $N = 4$ nothing slower than $t^{-3/2}$ is possible, whatever the topology and however many compartments vanish.

Fix $N = 4$, (G1)–(G2) and interior data, and order compartments topologically, so compartment 1 has no inflow and compartment 4 no outflow; y_4 never decreases and persists, so $\mathcal{V} \subset \{1, 2, 3\}$. By (2) every active flux contains a vanishing factor, so

$$\dot v = J(p)v + \mathfrak{N}_1(v), \qquad \dot p = G(p)v + \mathfrak{N}_2(v), \tag{19}$$

with $p = y_{\mathcal{P}}$, $v = y_{\mathcal{V}}$, matrices $J(p)$, $G(p)$ whose entries are linear forms in p ($J(p)$ Metzler, with $J(p^*) = J$ as in (25)), and quadratic forms $\mathfrak{N}_1, \mathfrak{N}_2$. The face $\{v = 0\}$ consists of equilibria. When $|\mathcal{P}| = 2$, say $\mathcal{P} = \{\ell, m\}$, write $\pi = p_\ell/(p_\ell + p_m)$; then $J(p) = (1 - |v|)J_{\mathrm{f}}(\pi)$ with J_{f} affine in π, where $|v| = \sum_{i\in\mathcal{V}} y_i$.

Lemma 5.5 (Diagonal lower bound)**.** *For $i \in \mathcal{V}$ there is C with $\dot y_i \ge y_i(J_{ii}(p) - C|v|)$, so $J_{ii}(p(t)) \ge 0$ cannot hold for all large t.*

Proof. Keep only the autocatalytic inflow $c_{aii}y_a y_i$ ($a \in \mathcal{P}$); outflow with vanishing catalysts is at most $C|v|y_i$. If $J_{ii}(p) \ge 0$ for $t \ge T$ then $y_i(t) \ge y_i(T)\exp(-C\int_T^\infty |v|) > 0$, since vanishing coordinates are integrable (Lemma 2.2). □

Lemma 5.6 (Desingularisation)**.** *Let $\xi > 0$ be C^1, $\xi \to 0$, $\int_0^\infty \xi < \infty$, and $\pi(t) \to \pi_\infty$. Let Ψ be real-analytic near π_∞ with $\Psi(\pi_\infty) = 0$, b Lipschitz there, and for large t*

$$\dot\xi = \xi(\Psi(\pi) + \rho_1), \qquad \dot\pi = \xi(b(\pi) + \rho_2), \qquad |\rho_1| + |\rho_2| \le C\xi.$$

Then $\Psi \not\equiv 0$, $b_\infty = b(\pi_\infty) \ne 0$, and with m the order of the zero of Ψ and $k = m + 1$, $C_\infty = -\Psi^{(m)}(\pi_\infty)(-b_\infty)^m/(k\,m!) > 0$ and $\xi(t) \sim C_\infty((k-1)C_\infty t)^{-k/(k-1)}$.

Proof. In exposure time $u = \int \xi$ (finite total U), $\mathrm{d}\xi/\,\mathrm{d}u = \Psi(\pi) + \rho_1$, $\mathrm{d}\pi/\,\mathrm{d}u = b(\pi) + \rho_2$. If $b_\infty = 0$, then $X = (\pi - \pi_\infty, \xi)$ obeys $|\,\mathrm{d}X/\,\mathrm{d}u| \le C|X|$, so $|X|$ stays bounded below on $[u_1, U)$, contradicting $X \to 0$. If $\Psi \equiv 0$, then $|\,\mathrm{d}\xi/\,\mathrm{d}u| \le C\xi$ and ξ is bounded below likewise. So $b_\infty \ne 0$ and $m < \infty$. Then $\pi - \pi_\infty = -(b_\infty + o(1))(U - u)$ and $\Psi(\pi(u)) = (L + o(1))(U - u)^m$ with $L = \Psi^{(m)}(\pi_\infty)(-b_\infty)^m/m!$. With $M(u) = \sup_{[u,U)} \xi$, $\xi(u) = \int_u^U(-\Psi - \rho_1)$ gives $M \le C_1(U - u)^{m+1} + C(U - u)M$, hence $M = O((U - u)^{m+1})$, $\int_u^U |\rho_1| = O((U - u)^{m+2})$ and $\xi = (-L/k + o(1))(U - u)^k$, so $C_\infty = -L/k > 0$. Finally $\tau = U - u$ obeys $\dot\tau = -(C_\infty + o(1))\tau^k$, which gives the stated law. □

Theorem 5.7 (Slowest tail at $N = 4$)**.** *With $N = 4$, (G1)–(G2) and interior data, every vanishing coordinate either decays exponentially or satisfies $y_i \sim C_i t^{-2}$ or $y_i \sim C_i t^{-3/2}$ with $C_i > 0$. In particular every vanishing coordinate is $O(t^{-3/2})$, and the exponent $3/2$ is attained (Theorem 3.1).*

Proof. *$|\mathcal{V}| = 1$:* Corollary 5.4.

$\mathcal{V} = \{1, 2, 3\}$: $\mathcal{P} = \{4\}$ and compartment 4 donates nothing, so the only off-diagonal entries of J come from a vanishing donor j with catalyst 4, which requires $j < i$. Thus J is lower triangular with diagonal $-h_i^* < 0$ (Lemma 2.2), and Theorem 7.1 gives exponential decay.

$\mathcal{V} = \{1, i\}$, $i \in \{2, 3\}$: y_1 has no inflow and hazard $h_1 \to h_1^* > 0$, so it decays exponentially; row 1 of J is $(-h_1^*, 0)$, so if $J_{ii} < 0$ all decay is exponential. Let $J_{ii} = 0$. Every inflow into i

not proportional to y_i contains y_1, so $\dot{y}_i = y_i(J_{ii}(p) + O(|v|)) + y_1 R$ with $R \geq 0$ bounded. With $4\omega = h_1^*$, eventually $y_1 \leq Ce^{-3\omega t}$, while $\dot{y}_i/y_i \geq J_{ii}(p) + O(|v|) \to 0$ gives $y_i \geq ce^{-\omega t}$; hence $y_1/y_i = O(y_i^2)$. So $\xi = y_i$, $\Psi(\pi) = J_{ii}(p_{\mathrm{f}}(\pi))$ and the face coordinate π satisfy Lemma 5.6. Ψ is affine and not identically zero, so $m = 1$: decay t^{-2}.

$\mathcal{V} = \{2, 3\}$*:* $\mathcal{P} = \{1, 4\}$ and $\pi = y_1/(y_1 + y_4)$ is nonincreasing because $\dot{y}_1 \leq 0 \leq \dot{y}_4$, so $\pi \downarrow \pi_\infty$. The affine functions $g_i(\pi) = J_{ii}(p_{\mathrm{f}}(\pi))$ satisfy $g_i(0) = -h_i(e_4) \leq 0$, since compartment 4 donates nothing. If $g_i(\pi_\infty) \geq 0$, then g_i is nondecreasing, so $g_i(\pi(t)) \geq 0$ for all t, contradicting Lemma 5.5. Hence $J_{22}, J_{33} < 0$. If J is reducible, or irreducible with $s(J) < 0$, decay is exponential. It remains to treat irreducible J with $\det J = 0$. Near p^*, the Perron root $\lambda(p)$ of $J(p)$ is simple and analytic, with positive analytic left and right eigenvectors $\psi(p)$, $\varsigma(p)$ ($\psi \cdot \varsigma = 1$); the other eigenvalue tends to $\operatorname{tr} J < 0$. Put $\xi = \psi(p) \cdot v \geq c|v|$. By (19), $\dot{\xi} = \xi(\lambda_{\mathrm{f}}(\pi) + \rho_1)$ with $|\rho_1| \leq C\xi$. The direction $q = v/\xi$ satisfies $\psi(p) \cdot q = 1$ and, by (19) with $\dot{p} = O(\xi)$, $\dot{q} = (J(p) - \lambda(p))q + O(\xi)$. Let $\lambda_2(p)$ be the other eigenvalue, with right eigenvector $\mathbf{r}_2(p)$ and left vector $\omega_2(p)$ normalised by $\omega_2 \cdot \mathbf{r}_2 = 1$, $\omega_2 \cdot \varsigma = 0$, and write $q = \varsigma(p) + e_\perp \mathbf{r}_2(p)$. Then $\dot{e}_\perp = (\lambda_2 - \lambda)e_\perp + O(\xi)$. Since $\lambda_2 - \lambda \to \operatorname{tr} J < 0$, there are $c > 0$ and T with $\lambda_2 - \lambda \leq -2c$ and $|\dot{\xi}/\xi| \leq c$ on $[T, \infty)$. Then $\xi(\sigma) \leq \xi(t)e^{c(t-\sigma)}$ for $T \leq \sigma \leq t$, and variation of constants gives $|e_\perp(t)| \leq e^{-2c(t-T)}|e_\perp(T)| + C\int_T^t e^{-2c(t-\sigma)}\xi(\sigma)\,\mathrm{d}\sigma \leq C'\xi(t)$, using also $e^{-2c(t-T)} \leq e^{-c(t-T)} \leq \xi(t)/\xi(T)$. Hence $|v/\xi - \varsigma(p)| \leq C\xi$. Hence $\dot{\pi} = \xi(b(\pi) + \rho_2)$ with b analytic and $|\rho_2| \leq C\xi$. Lemma 5.6 applies to $\Psi = \lambda_{\mathrm{f}}$. Near π_∞, $\det J_{\mathrm{f}} = \lambda_{\mathrm{f}} \cdot$(a negative function), so m is the order of the zero of $\det J_{\mathrm{f}}$, a nonzero polynomial of degree at most two. Thus $k \in \{2, 3\}$, and both coordinates follow the same law because $v \sim \xi\,\varsigma(p^*)$ with $\varsigma(p^*) > 0$. □

Remark 5.8 (Examples with two vanishing compartments; numerical). The model $\dot{y}_1 = -By_1y_3$, $\dot{y}_2 = By_1y_3 - y_2y_4 - y_2y_1$, $\dot{y}_3 = y_2y_4 - y_3y_4 - y_3y_1$, $\dot{y}_4 = y_2y_1 + y_3y_4 + y_3y_1$ satisfies (G1), and (G2) whenever $\mathcal{V} = \{2, 3\}$ (the only dependency cycle is $1 \to 3 \to 1$). On the face, $J_{\mathrm{f}}(\pi) = \left(\begin{smallmatrix} -1 & B\pi \\ 1-\pi & -1 \end{smallmatrix}\right)$ and $\lambda_{\mathrm{f}} = -1 + \sqrt{B\pi(1-\pi)}$. For $B = 4$ the root is double at $\pi = 1/2$, with predicted $t^{3/2}(y_2 + 2y_3) \to \sqrt{3}/2$; for $B = 9/2$ it is simple at $\pi = 2/3$, with predicted $t^2(y_2 + 3y_3) \to 16/3$. Near-critical interior data found by bisection give $0.806, 0.844, 0.859, 0.870$ and $5.07, 5.25, 5.31, 5.33$ at $t = 10^4, 10^{4.5}, 10^5, 10^{5.5}$ (package script `check_four_state.py`). Existence of interior data landing exactly on these zeros is supported numerically but not proved, and whether a t^{-2} tail occurs when $\mathcal{V} = \{1, i\}$ is open. The argument uses that the face of equilibria is at most a curve, which fails for $N \geq 5$.

6 The four-state chain: complete analysis

Goal: for the simplest counterexample topology, describe every trajectory, show that algebraic extinction is non-generic, quantify how quickly exponential decay returns, and show that the slow tails never follow an outbreak.

6.1 Classification

Consider the chain with arbitrary rates,

$$\dot{s} = -\alpha sx, \qquad \dot{w} = \alpha sx - \beta wx, \qquad \dot{x} = (\beta w - \gamma r)x, \qquad \dot{r} = \gamma rx, \tag{20}$$

$\alpha, \beta, \gamma > 0$, interior data. In exposure time $s = s_0e^{-\alpha u}$, $r = r_0e^{\gamma u}$, $w = w_0e^{-\beta u} + \frac{\alpha s_0}{\beta - \alpha}(e^{-\alpha u} - e^{-\beta u})$ for $\alpha \neq \beta$ and $w = e^{-\alpha u}(w_0 + \alpha s_0 u)$ for $\alpha = \beta$, and $\Phi = 1 - s - w - r$, with

$$\Phi' = \beta w - \gamma r, \quad \Phi'' = \alpha\beta s - \beta^2 w - \gamma^2 r, \quad \Phi''' = -\alpha\beta(\alpha + \beta)s + \beta^3 w - \gamma^3 r. \tag{21}$$

Theorem 6.1 (All rates and all interior data)**.** *Φ has a first positive zero U, $u(t) \uparrow U$, $x \to 0$, and s, w, r have positive limits. Exactly one of the following holds.*

(i) $\omega_* = \gamma r^* - \beta w^* > 0$*:* $x(t) \sim Ce^{-\omega_* t}$*,* $C > 0$*.*

(ii) $\beta w^* = \gamma r^*$ *and* $\kappa = \alpha\beta s^* - \gamma(\beta+\gamma) r^* > 0$*:* $x(t) \sim 2/(\kappa t^2)$*.*

(iii) $\beta w^* = \gamma r^*$ *and* $\kappa = 0$*: the endpoint is*

$$s^* = \frac{\gamma}{\alpha+\gamma}, \quad w^* = \frac{\alpha\gamma}{(\alpha+\gamma)(\beta+\gamma)}, \quad r^* = \frac{\alpha\beta}{(\alpha+\gamma)(\beta+\gamma)}, \tag{22}$$

and $x(t) \sim \frac{\sqrt{3}}{2\sqrt{\alpha\beta\gamma}}\, t^{-3/2}$*.*

No higher order occurs, and for every rate triple all three cases occur for suitable interior data.

Proof. $s, w, r > 0$ for finite u and $\Phi(0) = x_0 > 0$; eventually $r_0 e^{\gamma u} > 1$, so Φ has a first zero U. The solution of $\dot{u} = \Phi(u)$ increases to U without reaching it. Since $\Phi > 0$ on $[0, U)$, $\Phi'(U) \le 0$. Case (i) is a simple zero (Proposition 5.3, Step 4). If $\Phi'(U) = 0$ then $\Phi''(U) = \kappa \ge 0$; $\kappa > 0$ gives a double zero with $a_2 = \kappa/2$. If $\kappa = 0$, solving the two conditions with $s^* + w^* + r^* = 1$ gives (22), and substitution gives $\Phi'''(U) = -\alpha\beta\gamma < 0$, a triple zero with $a_3 = \alpha\beta\gamma/6$; this also excludes higher order. (16) gives the constants. For attainability, fix an endpoint on the face $x = 0$ with $\beta w^* < \gamma r^*$ (simple), with $\beta w^* = \gamma r^*$ and $0 < r^* < \alpha\beta/((\alpha+\gamma)(\beta+\gamma))$ (double), or (22) (triple), and integrate the linear exposure system backwards over a short interval; the sign of the leading Taylor term (for the triple case, $\Phi'''(U) < 0$) gives $\Phi > 0$ just before U, and continuity keeps the other fractions positive. □

For rates $(2, 1, 1)$ the profile is $\Phi = Q(q)/q$ with the cubic (4), and the attained root is the largest root $q_* \in (0, 1)$; a multiple root below a simple root that is reached first does not determine the tail. The double-zero data are exactly

$$s_0 = \frac{h}{q_*^2}, \quad w_0 = \frac{3h+1}{2q_*} - \frac{2h}{q_*^2}, \quad r_0 = \frac{q_*(1-h)}{2}, \qquad \frac{1}{3} < h < 1, \quad \frac{4h}{3h+1} < q_* < 1, \tag{23}$$

with $Q = s_0(q-q_*)^2(q-q_\circ)$, $q_\circ = q_*(1-h)/(2h)$, endpoint $(h, (1-h)/2, 0, (1-h)/2)$ and $x \sim \frac{2}{3h-1} t^{-2}$. Indeed, write a double root as $Q = s_0(q-q_*)^2(q-q_\circ)$. Then $q_\circ > 0$ because $r_0 > 0$; $q_\circ \ge 1$ contradicts $Q(1) = x_0 > 0$; $q_\circ \in (q_*, 1)$ would be attained first; and $q_\circ = q_*$ is the triple case. Hence $q_\circ < q_*$, which is $h > 1/3$, and the remaining inequality is $w_0 > 0$. Example (9) is $h = 1/2$, $q_* = 9/10$.

6.2 Genericity

Theorem 6.2 (Algebraic extinction is non-generic)**.** *Let* (15) *be a model on the open simplex* Ω *(dimension* $N-1$*), and let* $\Omega_0 \subset \Omega$ *be the set of data for which* x *is integrable (under (G1)–(G2), this is the set for which* x *vanishes). Let* $\Omega_{\exp} \subset \Omega_0$ *be the data for which* x *decays exponentially, and* E_k *the data whose first zero has order at least* k*. Then:*

(a) $\Omega_{\exp}$ *is open in* Ω*;*

(b) $E_2 = \Omega_0 \setminus \Omega_{\exp}$ *has Lebesgue measure zero, so* $\Omega_{\exp}$ *has full measure in* Ω_0*; if* Ω_0 *is open (for example* $\Omega_0 = \Omega$*),* $\Omega_{\exp}$ *is also dense in it;*

(c) E_k *is contained in the image under a smooth map of a manifold of dimension* $N - 1 - r_k$*, where* $r_k \le k-1$ *is the rank of the affine conditions* $e_x^{\mathsf{T}} H^j y^* = 0$ $(1 \le j < k)$ *on the face* $\{y_x = 0,\ \mathbf{1}^{\mathsf{T}} y = 1\}$*, so its codimension is at least* r_k*; the data of exact order* k *form, locally, an immersed submanifold of that dimension, and* E_k *is the finite union of these strata.*

For the four-state chain (20)*,* $\Omega_0 = \Omega$*, and at fixed rates the* t^{-2} *data form a surface (codimension one) and the* $t^{-3/2}$ *data a curve (codimension two).*

Proof. For $y_0 \in \Omega$ write $\Phi_{y_0}(u) = e_x^{\mathsf{T}} e^{Hu} y_0$, which is linear in y_0 and analytic in u. If Φ_{y_0} has a positive zero, $u(t)$ increases to the first one, U, and x is integrable; conversely integrability gives $U < \infty$ and $\Phi_{y_0}(U) = 0$. By Proposition 5.3 (Step 4, which uses only the form (15)), x decays exponentially iff the first zero is simple.

(a) Let U be a simple first zero of Φ_{y_0}. Choose $\eta > 0$ with $\Phi'_{y_0} < 0$ on $[U-\eta, U+\eta]$. For y_1 near y_0, Φ_{y_1} and Φ'_{y_1} are uniformly close to Φ_{y_0} and Φ'_{y_0} on $[0, U+\eta]$, so $\Phi_{y_1} > 0$ on $[0, U-\eta]$, $\Phi'_{y_1} < 0$ on $[U-\eta, U+\eta]$, and Φ_{y_1} changes sign there. Its first zero is therefore simple.

(b,c) Let Π be the affine face $\{y_x = 0, \mathbf{1}^{\mathsf{T}} y = 1\}$ (dimension $N-2$) and $\Lambda(y^*, U) = e^{-HU} y^*$ on $\Pi \times (0, \infty)$; since $\mathbf{1}^{\mathsf{T}} H = 0$, Λ maps into $\{\mathbf{1}^{\mathsf{T}} y = 1\}$. Every $y_0 \in \Omega_0$ equals $\Lambda(y(U), U)$, and the pair is determined by y_0. The first zero has order at least k iff in addition $\ell_j(y^*) := e_x^{\mathsf{T}} H^j y^* = \Phi^{(j)}(U) = 0$ for $1 \le j < k$. If ℓ_1 vanishes identically on Π then, since Π spans $\{y_x = 0\}$, $\ell_1(y) = H_{xx} y_x$ and $\Phi' = H_{xx}\Phi$, so $\Phi = x_0 e^{H_{xx} u}$ never vanishes and $\Omega_0 = \varnothing$. Otherwise the conditions cut out an affine subspace $\Pi_k \subset \Pi$ of dimension $N - 2 - r_k$ with $r_k \ge 1$ for $k \ge 2$, and $E_k \subset \Lambda(\Pi_k \times (0, \infty))$. The differential of Λ at (y^*, U) is $(\delta y, \delta U) \mapsto e^{-HU}(\delta y - H y^* \delta U)$. At a zero of exact order $k' \ge k$ the vector Hy^* satisfies $\ell_j(Hy^*) = \Phi^{(j+1)}(U)$, which vanishes for $j < k' - 1$ and not for $j = k' - 1$, while $e_x^{\mathsf{T}} H y^* = \Phi'(U) = 0$; so Hy^* is not tangent to $\Pi_{k'}$, and Λ restricted to $\Pi_{k'} \times (0, \infty)$ is an immersion at points of exact order k' (at points of higher order it need not be, which is why (c) is stated as a containment plus a stratification by exact order $k' \in \{k, \dots, N-1\}$). A C^1 image of a manifold of dimension $N-2$ in an $(N-1)$-dimensional space has measure zero, which proves the first part of (b), and the same argument with Π_k proves (c). If Ω_0 is open, the open set Ω_{exp} differs from it by a null set, which has empty interior, so Ω_{exp} is dense in Ω_0.

For the chain, Theorem 6.1 gives $\Omega_0 = \Omega$. The double-zero condition $\ell_1 = \beta w^* - \gamma r^* = 0$ is one affine condition on the two-dimensional face, and the triple endpoint (22) is a point, so $r_2 = 1$ and $r_3 = 2$. For rates $(2, 1, 1)$ this is also visible in the explicit parameterisations: the map $(h, q_*) \mapsto (s_0, w_0, r_0)$ in (23) has a 2×2 minor $-(3h-1)/(2q_*^2) \neq 0$, and the curve (5) has $\mathrm{d}s_0/\,\mathrm{d}q_* \neq 0$ (exact check in the package). □

Table 5 collects the resulting dictionary between the order of the zero, the tail, the amount of tuning, and the response to perturbation (Theorem 6.3).

order k of zero	tail of x	codimension (chain)	relaxation rate	endpoint shift
1	$e^{-\omega t}$	0 (open set)	—	$O(\varepsilon)$
2 (spinodal)	t^{-2}	1	$\propto \lvert\varepsilon\rvert^{1/2}$ one side	$\lvert\varepsilon\rvert^{1/2}$ or jump
3 (tricritical)	$t^{-3/2}$	2	$\propto \lvert\varepsilon\rvert^{2/3}$	$\propto \lvert\varepsilon\rvert^{1/3}$
k (class (15))	$t^{-k/(k-1)}$	$r_k \le k-1$	$\propto \lvert\varepsilon\rvert^{(k-1)/k}$	$\lvert\varepsilon\rvert^{1/k}$; jump if k even

Table 5: Order of the zero, tail, tuning and response to a perturbation of size ε.

6.3 How fast exponential decay returns

Theorem 6.3 (General-order slowing law)**.** *In the class (15), let y_0 have a first zero of order $k \ge 2$ at U, $\Phi(U - v) = a_k v^k + O(v^{k+1})$, $a_k > 0$. Let δ satisfy $\mathbf{1}^{\mathsf{T}}\delta = 0$, put $\Gamma(u) = e_x^{\mathsf{T}} e^{Hu}\delta$ and assume $g = \Gamma(U) \neq 0$. For small $\varepsilon \neq 0$ consider the data $y_0 + \varepsilon\delta$.*

(a) If k is odd, or k is even and $\varepsilon g < 0$, the first zero U_ε is simple, $U_\varepsilon - U = \mathrm{sgn}(\varepsilon g)\,\lvert\varepsilon g/a_k\rvert^{1/k}(1 + o(1))$, and x decays exponentially with rate

$$\omega_\varepsilon = k\, a_k^{1/k}\, \lvert\varepsilon g\rvert^{(k-1)/k}\,(1 + o(1)). \tag{24}$$

(b) If k is even and $\varepsilon g > 0$, there is $\eta > 0$, independent of ε, with $\Phi_\varepsilon > 0$ on $[0, U+\eta]$: the endpoint jumps by at least η in exposure.

In case (a) the endpoint moves by $y^*_\varepsilon - y^* = Hy^*\,\Delta U + O((\Delta U)^2 + |\varepsilon|)$, $\Delta U = U_\varepsilon - U$, with $Hy^* \neq 0$: a shift of order $|\varepsilon|^{1/k}$.

Proof. $\Phi_\varepsilon = \Phi + \varepsilon\Gamma$ by linearity. Fix a small $\eta > 0$ such that $\Phi' < 0$ on $[U-\eta, U)$, Φ has no zero in $(U, U+\eta]$ when k is even, and $|\Phi(U-v) - a_k v^k| \le a_k|v|^k/4$ for $|v| \le \eta$. On $[0, U-\eta]$, $\Phi \ge c > 0$, so $\Phi_\varepsilon > 0$ there for small ε. Substitute $v = |\varepsilon|^{1/k}\tau$. Uniformly for τ in compact sets, $|\varepsilon|^{-1}\Phi_\varepsilon(U-v) = a_k\tau^k + \operatorname{sgn}(\varepsilon)g + O(|\varepsilon|^{1/k})$. In case (a) the limit polynomial has exactly one real root $\tau_0 = -\operatorname{sgn}(\varepsilon g)|g/a_k|^{1/k}$ on the relevant side, and it is simple. When k is even and $\varepsilon g < 0$ the other root $-\tau_0$ lies beyond τ_0 in u. The implicit function theorem gives a simple zero $\tau_\varepsilon \to \tau_0$, and positivity of $a_k\tau^k + \operatorname{sgn}(\varepsilon)g$ before τ_0, together with the bound on $[U-\eta, U-|\varepsilon|^{1/k}T]$ for large T, shows that it is the first zero. Then $-\Phi'_\varepsilon(U_\varepsilon) = ka_k v_\varepsilon^{k-1}(1+o(1)) + O(\varepsilon)$ with $|v_\varepsilon| \sim |\varepsilon g/a_k|^{1/k}$, and $\varepsilon = o(|\varepsilon|^{(k-1)/k})$, which gives (24); the rate of x is $-\Phi'_\varepsilon$ at its simple first zero (Proposition 5.3, Step 4). In case (b), $\Phi_\varepsilon \ge \Phi + \varepsilon\min\Gamma > 0$ on $[U-\eta, U+\eta]$ once ε is small, since $\Phi \ge 0$ there and Γ has the sign of g near U. Finally $y^*_\varepsilon = e^{HU_\varepsilon}(y_0 + \varepsilon\delta) = e^{H\Delta U}y^* + \varepsilon e^{HU_\varepsilon}\delta = y^* + Hy^*\Delta U + O((\Delta U)^2 + |\varepsilon|)$, and $Hy^* = 0$ would make y^* an equilibrium of the exposure flow reached at finite U, forcing $\Phi \equiv 0$. □

For the base chain with $x_0 \mapsto x_0 + \varepsilon$, $r_0 \mapsto r_0 - \varepsilon$ we have $a_3 = 1/3$, $g = 5/4$, and (24) reproduces the exact rate $3(25/48)^{1/3}|\varepsilon|^{2/3}$, which here holds for every admissible $\varepsilon \neq 0$ because $Q_\varepsilon = \frac{25}{48}(q - \frac{4}{5})^3 + \varepsilon$. For the double-zero example (9) with the same perturbation, $\varepsilon < 0$ gives a rate $\approx 2 \cdot \frac{1}{2}|\frac{10}{9}\varepsilon|^{1/2}$, while $\varepsilon > 0$ moves the attained root from 9/10 to 9/20: the final stifler fraction jumps from 1/4 to 1/2 and the final unaware fraction from 1/2 to 1/8. This discontinuity of the final state across the codimension-one double-zero surface is a spinodal; the triple-zero curve is where the jump closes, a tricritical point in the sense used for generalised epidemics [9, 11, 14] and for the discontinuous transitions of complex contagion [10, 15]. The package checks (24) at high precision for these two examples and for the order-five zero of the $m = 4$ hierarchy. The relaxation time $\omega_\varepsilon^{-1} \propto |\varepsilon|^{-(k-1)/k}$ is a critical-slowing-down law of the kind used as an early-warning indicator of elimination [30].

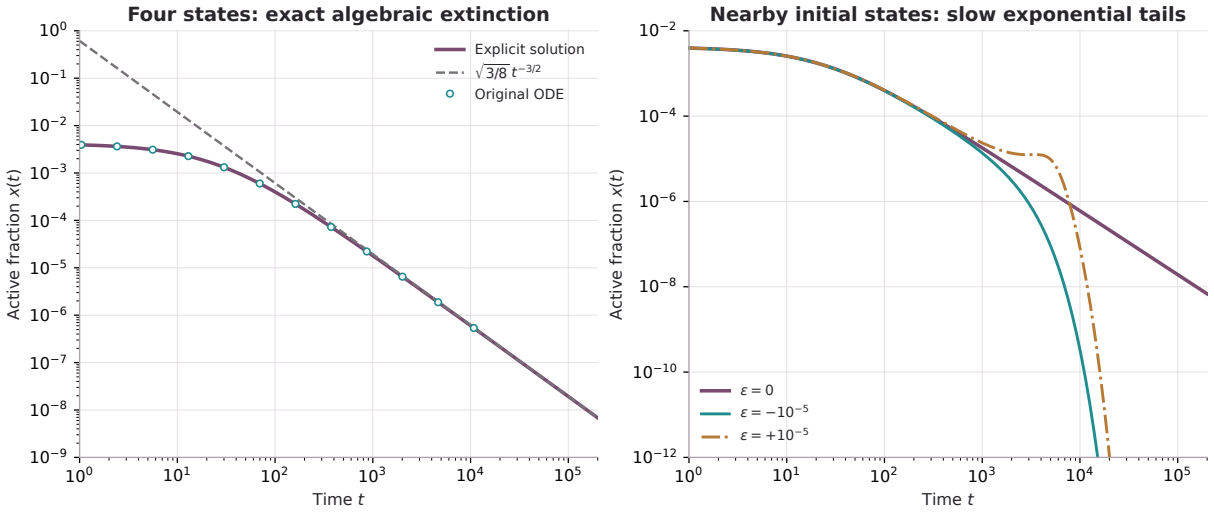


Figure 3: Left: the explicit trajectory of Theorem 3.1, a numerical integration of (3), and the asymptote $\sqrt{3/8}\,t^{-3/2}$. Right: perturbations $\varepsilon = \pm 10^{-5}$ of (x_0, r_0); both decay exponentially at the rate of Theorem 6.3. Illustration only.

6.4 No outbreak precedes a slow tail

Lemma 6.4 (Zeros of exponential polynomials)**.** *Let $f(u) = \sum_{j=1}^{n} p_j(u)e^{\lambda_j u}$ with distinct real λ_j and nonzero real polynomials p_j of degrees d_j. Then f has at most $D-1$ real zeros counted with multiplicity, where $D = \sum_j (d_j + 1)$.*

Proof. Induction on D; $D = 1$ is clear. Multiply by $e^{-\lambda_n u}$ and differentiate $d_n + 1$ times. This removes the n-th term and leaves an exponential polynomial with the other $\lambda_j - \lambda_n \neq 0$ and the same degrees, which has at most $D - d_n - 2$ zeros by induction. Rolle's theorem, counted with multiplicity, gives at most $(D - d_n - 2) + (d_n + 1) = D - 1$ zeros for f. (This is a classical generalisation of Descartes' rule; see [32].) □

Proposition 6.5 (Monotone fade-out)**.** *In the following cases $x(t)$ is strictly decreasing on $[0, \infty)$, so the largest activity is the initial one:*

(a) the chain (20), any rates, whenever the first zero is double or triple;

(b) SWIR (10), any $\nu, \eta > 0$, whenever the first zero is double or triple;

(c) the hierarchy (13), every even m.

Proof. Since $\dot{x} = \Phi'(u)x$, it suffices that $\Phi' < 0$ on $[0, U)$. (a) $\Phi' = \beta w - \gamma r$ is an exponential polynomial with $D = 3$ (exponents $-\alpha, -\beta, \gamma$; or $-\alpha$ with degree one and γ when $\alpha = \beta$), so it has at most two real zeros, and $\Phi'(u) \to -\infty$ as $u \to \infty$. At a triple zero, U is a double zero of Φ', so Φ' has no other zero, and it is negative near U. At a double zero, $\Phi'(U) = 0 < \Phi''(U)$, so Φ' crosses upwards at U; since it ends negative, it has a second zero after U, hence none before, and it is negative just before U. In both cases $\Phi' < 0$ on $(-\infty, U)$. (b) $\Phi' = \eta w - 1$ has exponents $-\nu, -\eta, 0$ (or degree one at $-\eta$ if $\nu = \eta$), so $D = 3$, and $\Phi' \to -1$; the argument of (a) applies. (c) From (13), $w_1 = \varrho\zeta \sum_{\ell=0}^{m-1}(\ell + 1)(-\vartheta)^\ell = \varrho(1 - (m + 1)\vartheta^m - m\vartheta^{m+1})/\zeta$, so $w_1 - r = -\varrho\vartheta^m(m + 1 + m\vartheta)/\zeta < 0$. □

For the base trajectory this is explicit: $w - r = -(\zeta - 1)^2(2\zeta + 1)/(3\zeta) < 0$. So in these families the slow tail is a slow fade-out from a subcritical start ($R_{\text{eff}} < 1$ throughout), not the end of an epidemic wave. Whether some model in the class has an algebraic tail after a genuine peak is open (Section 9).

7 What replaces the conjecture

Goal: identify exactly what goes wrong, and give conditions, checkable from the coefficient supports or the rates, under which the conjectured conclusion is true.

Assume (G1), let y^* be the limit, $\mathcal{V}$ and $\mathcal{P}$ the vanishing and persistent sets, $v = y_{\mathcal{V}}$, and assume $\mathcal{V} \neq \varnothing$. Let J be the Jacobian of the vanishing right-hand sides with respect to v at $(y^*_{\mathcal{P}}, 0)$. Collecting the terms of (1) that are linear in v at the limit, and using (2), for $i \neq j$ in $\mathcal{V}$

$$J_{ii} = \sum_{a \in \mathcal{P}} c_{aii} y^*_a - \sum_{k \in \mathcal{P}} h_{ik} y^*_k, \qquad J_{ij} = \sum_{k \in \mathcal{P}} c_{jik} y^*_k + \sum_{a \in \mathcal{P}} c_{aij} y^*_a, \qquad h_{ik} = \sum_{b \neq i} c_{ibk}. \tag{25}$$

The first sum in J_{ii} is *autocatalytic inflow*: a persistent donor moving into i on contact with i itself. J is Metzler.

Theorem 7.1 (Logarithmic decay rate from the limiting block)**.** *Under (G1), with interior data,*

$$\lim_{t \to \infty} \frac{1}{t} \log \|v(t)\|_1 = s(J) \leq 0. \tag{26}$$

Every vanishing coordinate has an exponential bound iff $s(J) < 0$, and then $\|y(t) - y^*\|_1 \le C_\omega e^{-\omega t}$ for every $0 < \omega < -s(J)$.

Proof. Step 1. Every nonzero flux has a factor in $\mathcal{V}$ by (2). Write each inflow into a vanishing coordinate as a nonnegative coefficient times one vanishing factor (if both factors vanish, choose one; if the catalyst is the destination itself, the coefficient goes on the diagonal), and put each outflow on the donor's diagonal. This gives $\dot v = M(t)v$ with $M(t)$ Metzler, since only diagonal entries can be negative, and $M(t) \to J$; terms with two vanishing factors contribute coefficients tending to zero.

Step 2 (upper bound). For $\mu > s(J)$, $\theta = (\mu I - J^\mathsf{T})^{-1}\mathbf{1} = \int_0^\infty e^{(J^\mathsf{T}-\mu)\sigma}\mathbf{1}\,\mathrm{d}\sigma > 0$ and $J^\mathsf{T}\theta = \mu\theta - \mathbf{1}$, so eventually $M(t)^\mathsf{T}\theta \le \mu\theta$ and $W = \theta^\mathsf{T}v$ satisfies $\dot W \le \mu W$. Hence $\limsup t^{-1}\log\|v\|_1 \le \mu$.

Step 3 (lower bound). Choose an irreducible diagonal block B of the Frobenius form of J with $s(B) = s(J)$, and a positive left eigenvector ψ (Perron–Frobenius [22]). Off-block terms are nonnegative, so $\dot v_B \ge M_B(t)v_B$, and $Z = \psi^\mathsf{T}v_B > 0$ satisfies $\dot Z \ge (s(J) - \epsilon)Z$ eventually, for every $\epsilon > 0$. Since $v \to 0$, $s(J) \le 0$.

Step 4. If $s(J) < 0$, Step 2 with $\mu = -\omega$ gives $\|v\|_1 \le Ce^{-\omega t}$; each flux has a vanishing factor, so $\|\dot y_\mathcal{P}\|_1 \le C'\|v\|_1$, and integrating from t to ∞ gives the bound on $y - y^*$. The converse follows from (26). □

The same result, with a different proof, is Theorem 7 and Corollary 8 of [2]. It is an a posteriori criterion: it needs the limit y^*, uses only (G1), and when $s(J) = 0$ it does not determine the slower rate. Step 2 is the standard linear-functional test for Metzler stability [23, 21]. For the base chain and SWIR, $J = [0]$ at the algebraic endpoints. The structure of (25) is what yields criteria without the limit.

Definition 7.2. The *limiting influence graph* on $\mathcal{V}$ has an edge $j \to i$ ($i \ne j$) when $c_{jik} > 0$ for some $k \in \mathcal{P}$ or $c_{aij} > 0$ for some $a \in \mathcal{P}$. By (25) and $y^*_\mathcal{P} > 0$, this is exactly the off-diagonal support of J.

Corollary 7.3 (A support-checkable repair)**.** *Assume (G1)–(G2), interior data, an acyclic limiting influence graph, and no autocatalytic inflow into vanishing states from persistent donors ($c_{aii} = 0$ for $a \in \mathcal{P}$, $i \in \mathcal{V}$). Then every vanishing state decays exponentially, and $y \to y^*$ exponentially.*

Proof. Acyclicity makes J permutation-similar to a triangular matrix, so $s(J) = \max_i J_{ii}$. Without autocatalytic inflow, $J_{ii} = -\sum_k h_{ik}y^*_k$. By Lemma 2.2 i has a persistent dependency successor k, so $h_{ik} > 0$ and $J_{ii} < 0$. Apply Theorem 7.1. □

The hypotheses of Corollary 7.3 are conditions on the supports of the c_{ijk} together with the labels $\mathcal{V}, \mathcal{P}$. They are checkable from the graphs alone when the labels are: by the source's Corollary 1 [1, p. 22], this holds under (G1)–(G2) "if the persistent successor within each cycle of the dependency graph is identifiable" (for example, when the dependency graph is acyclic). Without the acyclicity hypothesis, $s(J) = \max_i J_{ii}$ can fail and the triangular argument does not apply. The base chain violates exactly the autocatalysis condition ($W \to X$ catalysed by X), and so does SWIR. In the source's language, the dependency graph records the catalysts of *outflows* only; the counterexamples live in the catalysts of *inflows*. Whether this is the "edge case" mentioned in [1, p. 32] we cannot say.

Corollary 7.4 (A rate-checkable repair)**.** *Assume (G1)–(G2), interior data and an acyclic limiting influence graph. If for every $i \in \mathcal{V}$*

$$c_{kii} \le h_{ik} \text{ for all } k \in \mathcal{P}, \text{ with strict inequality for at least one } k \in \mathcal{P}, \tag{27}$$

then every vanishing state decays exponentially. No criterion that depends only on the rates can decide, without the initial data, whether an individual trajectory decays exponentially or algebraically: for the chain (20)*, every rate triple admits both.*

Proof. By (25), $J_{ii} = \sum_{k\in\mathcal{P}}(c_{kii} - h_{ik})y_k^* < 0$ because all $y_k^* > 0$; conclude as in Corollary 7.3. The second statement is the attainability part of Theorem 6.1. □

Condition (27) says that each persistent compartment helps remove i at least as strongly as it feeds i in i's presence. In the base chain it fails for $k = W$ ($c_{WXX} = 1 > h_{XW} = 0$). In SWIR with recovery encoded as in the source, $c_{WXX} = \eta$ and $h_{XW} = 1$, so (27) holds iff $\eta \le 1$; then SWIR always fades out exponentially, consistent with the triple-zero condition $w^* = 1/\eta < 1$. For given rates *and* data in the class (15), the test is the multiplicity of the first zero of the explicit function Φ; for rates $(2, 1, 1)$ it reduces to whether the discriminant of Q vanishes at the attained root, a finite exact computation.

8 Finite populations (exploratory)

Goal: indicate, without proof, what the deterministic tail means for a population of finite size.

The deterministic model is the large-population limit of a Markov chain [33]. We simulated that chain for the base model, with N individuals, events $S \to W$, $W \to X$, $X \to R$ at rates $2SX/N$, WX/N, RX/N and initial counts in the exact proportions (6), by Gillespie's method [31] with fixed seeds. Table 6 compares the median time at which the last spreader stops with the deterministic time $t_1(N)$ at which $Nx(t) = 1$.

N	runs	median T	10–90% of T	$t_1(N)$	median final r
24,000	200	98	51–233	556	0.298
240,000	60	283	151–552	2,718	0.311
2,400,000	20	986	411–1,658	12,810	0.345
24,000,000	20	1,676	870–2,914	59,777	0.319

Table 6: Exploratory simulation of the base model (package script `finite_population_gillespie.py`).

The median extinction time fits $1.48\,N^{0.42}$ (bootstrap 95% interval for the exponent $[0.38, 0.46]$), whereas $t_1(N)$ grows like $N^{2/3}$. A branching-process heuristic predicts $N^{2/5}$: a near-critical population of n spreaders dies out on a time scale of order n, and the deterministic count $Nx(t) \sim Nt^{-3/2}$ falls to order t once $t \gtrsim N^{2/5}$. At these sizes the deterministic tail overestimates the time to extinction by a factor of 6 to 36. Fluctuations of order $N^{-1/2}$ in the realised fractions act like the perturbation ε of Theorem 6.3, which would give relaxation times of order $N^{1/3}$; the final fractions also fluctuate, as expected near a tricritical point. Near-critical extinction and final-size scaling in stochastic SIR models is classical [29, 28]. For eradication timing this means that the deterministic algebraic tail is relevant only for very large populations; a proper stochastic analysis is outside the scope of this paper.

9 Discussion, related work and open problems

Goal: place the results and say what remains open.

Summary. As shown in [2] and here, the conjecture fails because a vanishing compartment can be fed, at a positive limiting rate, by a persistent compartment whose move is catalysed by the vanishing compartment itself, so the limiting growth rate J_{ii} can be exactly zero. In epidemiological terms the endgame lands on $R_{\text{eff}} = 1$. Within the single-catalyst class (15) this requires tuned data (Theorem 6.2), and away from the tuned set the decay is exponential, with a rate that returns as $|\varepsilon|^{(k-1)/k}$. Graph and rate conditions that exclude the mechanism restore the conclusion (Corollaries 7.3 and 7.4).

Relation to the prior release [2]. That release is an earlier Evidence Press publication, so this is a citation within the same publisher's series; it already refutes Conjecture 1, and the present paper builds on it. Overlaps and differences are as follows. *Counterexample:* its $18/t^2$ example has a cyclic dependency graph and is explicit in its internal clock; ours has both graphs acyclic, decays like $t^{-3/2}$ and is explicit in t. *Minimality:* the same statement (its Theorem 5); our proof differs in the final case but adds nothing to the statement. *Hierarchy:* its family has, with $N = m + 2$ compartments, exponent $(N-2)/(N-3)$ for every $N \geq 4$ and cyclic dependency graphs; ours has both graphs acyclic and exponent $(N-1)/(N-2)$ at even N, which is slower, and coincides with its exponent at odd N after the terminal split. Both families have the form (15), so Proposition 5.3 shows that ours is optimal in that class. *Classification:* it classifies its own four-compartment model, including the case in which the active compartment persists; ours classifies a different model, the chain (20), for arbitrary rates. *Spectral criterion:* its Theorem 7 and Corollary 8 state the same result as our Theorem 7.1, including reducible blocks and the aggregate logarithmic rate; our proof is a variant, and we claim no novelty for it. The remaining results (Theorems 5.7, 6.2, 6.3, Proposition 6.5, Corollaries 7.3 and 7.4, Section 8) have no counterpart there.

Related work. Table 7 compares the models discussed. Algebraic decay at non-hyperbolic equilibria is textbook centre-manifold behaviour [20]; the point here is that it is compatible with the source's graph conditions. The exponent 3/2 is the mean-field tricritical exponent of generalised epidemic processes [8, 9, 13], also studied on networks [12], whose discontinuous transitions are studied in [10, 11, 15, 14]; slow decay from spatial inhomogeneity is a different mechanism [26]. The models are conservative networks of catalysed conversions $Y_i + Y_k \to Y_j + Y_k$ with mass-action kinetics [16, 17]; persistence of such networks is studied through siphons [18], and their boundary equilibria are central to global-stability results [19]. The vanishing sets here are natural siphon candidates, and restating (G1)–(G2) in that language is left open. The exposure clock is classical [3, 24, 25]. Quadratisation of higher-degree systems [27] could produce high-order zeros in general, but need not respect positivity, conservation or the graph conditions; the hierarchy does. We make no claim of historical priority. A 2026 Leiden doctoral dissertation by the first author of [1] may contain a later version of the conjecture; we could not access its text, and only version 1 of [1] exists on arXiv.

model	states	trans. DAG	dep. DAG	in class, (G2)	slowest tail
SIR [3]	3	yes	no (self-loop)	yes	exponential
rumour scotching [24]	3	yes	yes	yes	exponential
prior example [2]	4	yes	no	yes	t^{-2}
prior family [2]	$m+2$	yes	no	yes	$t^{-m/(m-1)}$
SWIR-type, §3.4	4	yes	no	yes	$t^{-3/2}$
base chain (Thm 3.1)	4	yes	yes	yes	$t^{-3/2}$, explicit
hierarchy (Thm 5.1)	$m+2$	yes	yes	yes	$t^{-1-1/m}$

Table 7: Models discussed. "Slowest tail" is over interior data at the stated structure. The three-state rows follow from Theorem 4.1.

Open problems. (1) The slowest tail at fixed $N \geq 5$ outside the class (15), and whether t^{-2} occurs with $\mathcal{V} = \{1, i\}$ at $N = 4$ (Remark 5.8). (2) Genericity outside the single-catalyst class: is algebraic extinction always confined to a null set of data under (G1)–(G2)? (3) Is there any model in the class with an algebraic tail after a genuine outbreak peak? (4) A stochastic theory of the near-tricritical endgame (Section 8). (5) A siphon formulation of Corollary 7.3.

Scope. The results concern a deterministic model class; no data are analysed. The critical data are tuned, and the hierarchy is not proposed as a calibrated model.

Verification, data and code availability

The package accompanying this paper contains: an exact rational checker for the hierarchy, $m = 2, 4, \ldots, 20$ (`check_chain_family.py`); symbolic and high-precision checks of the derivation in Section 3, the SWIR example, the codimension computations, the monotonicity identities, the dominance examples and the slowing law (`check_extensions.py`); the four-state checks for Theorem 5.7 and Remark 5.8 (`check_four_state.py`); numerical integration of the original equations and the figures (`numerical_checks_and_figures.py`); and the exploratory simulation (`finite_population_gillespie.py`). A single script, `verify_all.py`, reruns all of them and compares every output byte for byte with the archived copy. These computations test the statements and their implementation; the general statements rest on the proofs. No proof assistant has checked the proofs. The package is archived as version 1.0.0-candidate at https://doi.org/10.5281/zenodo.23548109.

Use of AI tools

AI systems (large language models) were used extensively in this work: for mathematical exploration, drafting and revising the text, writing the verification code, and checking references. The author is responsible for the content. All claims labelled *exact* or *numerical* are reproduced by the accompanying scripts, and all references were checked against DOI or arXiv records.

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